

ZERO DIVISORS OF THE INTEGRAL SEDENIONS AND THE COMPACT G_2 PART IV OF THE OCTONION PROGRAM RECORD

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ABSTRACT. We study the zero divisors of the integral sedenions, the Cayley–Dickson double of Coxeter’s order of integral octonions, and their interaction with the arithmetic of the compact group G_2 . Among the 57,600 sedenions whose two octonion halves are units, exactly 7,560 are zero divisors, namely those whose halves are imaginary and orthogonal; they fall into two $G_2(\mathbb{Z})$ -orbits of sizes 6,048 and 1,512, distinguished by the type (Lipschitz or Hurwitz) of the quaternion subalgebra they generate, and the 84,672 ordered pairs (x, y) with $xy = 0$ form seven free $G_2(\mathbb{Z})$ -orbits: the integral shadow of Moreno’s theorem that the real unit zero-divisor pairs form the Lie group G_2 . Over $\mathbb{R}$ the zero-divisor manifold is $G_2/\mathrm{Sp}(1)$ and its Hecke algebra $\mathcal{H}(G_2, \mathrm{Sp}(1))$ is computed in low weights. The conic of a quaternion subalgebra modulo p meets the maximal-parahoric class set of the compact G_2 at p in a pattern governed by the octahedral group S_4 : for $p \in \{3, 5, 7, 11\}$ each conic lies in a single class determined by the type of the zero divisor, and for $p \geq 13$ it splits into S_4 -orbits, with class stabilisers $192/m$ and $48/m$; this is proved modulo one lemma verified for $p \leq 19$. Zero-divisor periods of algebraic modular forms on G_2 at level 3 are defined; they are the D^1 -periods of Gan and Savin’s exceptional see-saw, and the period along the conic separates Iwahori-new forms from old ones. Three constructions of a sedenion spectral geometry in the sense of Farnsworth and Boyle are shown to admit no finite Dirac operator beyond a scalar mass, and the Euler product of the norm-counting zeta function is shown to survive exactly up to the sedenions along the Cayley–Dickson ladder. All computations and code are deposited in the program’s repository.

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1. INTRODUCTION AND SUMMARY

Parts I and II of this record [1, 2] concern class sets and Hecke modules of the compact groups G_2 and F_4 built from Coxeter's integral octonions $\mathbb{O}_{\mathbb{Z}}$ (the E_8 lattice). The present part goes one rung further on the Cayley–Dickson ladder, to the sedenions $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$, and asks what the first failure of division has to do with the arithmetic developed so far. The answer is more structured than expected.

- (a) The integral sedenions $\mathbb{S}_{\mathbb{Z}} = \mathbb{O}_{\mathbb{Z}} \oplus \mathbb{O}_{\mathbb{Z}}\ell$ form the lattice $E_8 \oplus E_8$. Their zero divisors with unit halves are *exactly* the elements $a + b\ell$ with a, b imaginary orthogonal units (Theorem 2.1); there are 7,560 of them, each with a 4-dimensional annihilator, and they form two $G_2(\mathbb{Z})$ -orbits separated by the type of the quaternion subalgebra $\langle 1, a, b, ab \rangle$. The ordered zero-divisor pairs form seven free $G_2(\mathbb{Z})$ -orbits (Theorem 2.2).
- (b) Over $\mathbb{R}$ the unit zero divisors form $G_2/\mathrm{Sp}(1)$ and Moreno's G_2 is the sphere bundle of their annihilators. The Hecke algebra $\mathcal{H}(G_2, \mathrm{Sp}(1))$, a non-commutative algebra with continuous families of idempotents, has blocks of sizes 1, 3, 3, 6, 8, 10, 6 on the representations of dimensions 1, 7, 14, 27, 64, 77, 77' (Section 3).
- (c) The conic law (Theorem 4.2): the isotropic lines of $\mathrm{Im} \mathbb{H}_z \otimes \mathbb{F}_p$ form a conic $\cong \mathbb{P}^1(\mathbb{F}_p)$, and its intersection with the maximal-parahoric class set of the compact G_2 at p is governed by the octahedral group S_4 . The two zero-divisor types index two classes for $p = 3, 5, 7, 11$ and the conics split into cube-orbits for $p \geq 13$.
- (d) Zero-divisor periods of algebraic modular forms (Section 5) are the D^1 -periods of the Gan–Savin see-saw [5]; the period along the conic vanishes exactly on the Iwahori-new forms of weight $2\omega_1$ at level 3, while their periods on other orbits do not.
- (e) Negatives (Section 6): in the Farnsworth–Boyle framework of non-associative spectral geometry [7, 8] the sedenions admit, in all constructions tried, no finite Dirac operator beyond a scalar mass; the sedenion analogue of the Albert algebra is not Jordan; and the Euler product of the norm-counting zeta function dies one doubling past the sedenions, where Ramanujan's Δ first enters.

Everything in (a)–(d) was computed exactly (integer arithmetic, finite groups, or linear algebra at machine precision with independent checks), and the one statement not fully proved, the general case of the conic law, is marked as such.

2. THE INTEGRAL SEDENIONS AND THEIR ZERO DIVISORS

Let $\mathbb{O}_{\mathbb{Z}}$ be Coxeter's maximal order of integral octonions, with norm N , trace tr and the 240 units. The Cayley–Dickson double $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$ has product $(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c})$, conjugation $\overline{(a, b)} = (\bar{a}, -b)$ and norm $N(a, b) = N(a) + N(b)$; we take $\mathbb{S}_{\mathbb{Z}} = \mathbb{O}_{\mathbb{Z}} \oplus \mathbb{O}_{\mathbb{Z}}\ell$. On 200 random triples the laws behave as classically known: $\mathbb{S}$ is flexible and power-associative; it is not commutative, associative or alternative, and $N(xy)/(N(x)N(y))$ ranged over $[0.709, 1.316]$.

Theorem 2.1. *Let $x = (a, b) \in \mathbb{S}_{\mathbb{Z}}$ with $N(a) = N(b) = 1$. Then x is a zero divisor if and only if a and b are both imaginary ($\mathrm{tr} = 0$) and orthogonal. There are 7,560 such elements; each has a 4-dimensional annihilator $\{y : xy = 0\}$. No unit of $\mathbb{S}_{\mathbb{Z}}$ (norm 1) is a zero divisor.*

Proof. Exhaustive computation over all $240^2 = 57,600$ pairs (a, b) of units: the left-multiplication matrix L_x is singular exactly for the 7,560 pairs with $\mathrm{tr} a = \mathrm{tr} b = 0$ and $\langle a, b \rangle = 0$ (the other seven combinations of the three conditions never give a zero divisor), and its kernel has dimension 4 in every case. A unit has one half zero and all 230,400 products of units have norm 1. $\square$

The real version (imaginary, orthogonal, equal norm) is due to Moreno and to Khalil–Yiu; the count and the dimension 4 reproduce Moreno’s dimension count $11 + 3 = 14 = \dim G_2$ for the set of unit zero-divisor pairs [3].

Theorem 2.2. *The group $G_2(\mathbb{Z}) = \text{Aut}(\mathbb{O}_{\mathbb{Z}})$ (order 12,096) acts on the 7,560 zero divisors with two orbits, of sizes 6,048 and 1,512, with stabilisers of orders 2 and 8. The quaternion subalgebra $\mathbb{H}_z = \langle 1, a, b, ab \rangle \cap \mathbb{O}_{\mathbb{Z}}$ contains 8 units (Lipschitz type) on the large orbit and 24 (Hurwitz type) on the small one, and these are exactly the numbers of zero divisors y in the annihilator of x among the 7,560. The $84,672 = 7 \cdot 12,096$ ordered pairs (x, y) of these zero divisors with $xy = 0$ form seven free $G_2(\mathbb{Z})$ -orbits.*

Proof. Computation with the 12,096 automorphisms in the Coxeter basis (each verified to be a sedenion automorphism acting diagonally on the two halves); the orbit and stabiliser counts are exact. The map $(a, b) \mapsto (a, -b)$ is an automorphism of $\mathbb{S}$; the half-swap preserves the zero-divisor set but is not an automorphism. $\square$

Remark 2.3. Moreno’s theorem says the real unit zero-divisor pairs form a space homeomorphic to G_2 ; Theorem 2.2 is its integral shadow: seven copies of $G_2(\mathbb{Z})$, acted on freely.

3. THE ZERO-DIVISOR MANIFOLD AND ITS HECKE ALGEBRA

Over $\mathbb{R}$, a unit zero divisor is a pair (a, b) of orthonormal imaginary octonions. G_2 acts transitively on such pairs, so the zero-divisor manifold is $ZD = G_2/K$ with K the stabiliser of one pair.

Proposition 3.1. *K is 3-dimensional with Lie algebra $\mathfrak{su}(2)$; it fixes $\mathbb{H}_z$ pointwise and acts on the 4-dimensional annihilator of $z = (a, b)$ as $\text{Sp}(1)$ acts on $\mathbb{H}$ (the commutant of K on the annihilator is 4-dimensional). Hence $ZD = G_2/\text{Sp}(1)$ is the Stiefel manifold of orthonormal pairs in $\mathbb{R}^7$, of dimension 11, and Moreno’s G_2 is the unit-sphere bundle of the annihilators over it.*

Proof. Computed from the 14 derivations of $\mathbb{O}$ in an orthonormal basis: the derivations killing a and b form a 3-dimensional subalgebra, which kills $\langle 1, a, b, ab \rangle$ and preserves the annihilator, on which its commutant has dimension 4. $\square$

In Gan–Savin’s notation [5], $K = D^1$ is the group of norm-one elements of the quaternion subalgebra $D = \mathbb{H}_z$, acting by $g \cdot (a, b) = (a, b\bar{g})$; it is the pointwise stabiliser of D in $\text{Aut}(\mathbb{O})$.

The Hecke algebra of the homogeneous space is $\mathcal{H}(G_2, K) = \bigoplus_{\pi} M_{m_{\pi}}(\mathbb{C})$ with $m_{\pi} = \dim \pi^K$ (Frobenius reciprocity: m_{π} is the multiplicity of π in $L^2(ZD)$). Decomposing $7 \otimes 7$, $7 \otimes 14$, $\text{Sym}^3 7$ and $\text{Sym}^2 14$ by the Casimir and counting K -fixed vectors:

π	1	7	14	27	64	77	77'
$m_{\pi} = \dim \pi^K$	1	3	3	6	8	10	6

The 7 gives 3 because K fixes exactly $\text{Im } \mathbb{H}_z \subset \text{Im } \mathbb{O}$; the 14 gives 3 because the centraliser of K in $\mathfrak{g}_2$ is the other $\mathfrak{su}(2)$ of $\mathfrak{so}(4) \subset \mathfrak{g}_2$. Since (G_2, K) is not a Gelfand pair, the idempotents of each block form continuous families (Grassmannians). The representation theory here is classical; the identification of ZD with the sedenion zero divisors is what makes it ours. Every weight in the Path A table of compact G_2 forms (Part II) has $m_{\pi} > 0$, which is what makes the periods of Section 5 possible.

4. QUATERNION SUBALGEBRAS AND THE CONIC LAW

4.1. Quaternion subalgebras of the Coxeter order. Every zero divisor $z = (a, b)$ determines $\mathbb{H}_z = \langle 1, a, b, ab \rangle \cap \mathbb{O}_{\mathbb{Z}}$, a quaternion order of the definite quaternion algebra it spans.

Counting ordered orthogonal imaginary unit pairs per subalgebra ($6 \cdot 4 = 24$ in both cases) gives:

Proposition 4.1. $\mathbb{O}_{\mathbb{Z}}$ contains 63 quaternion subalgebras of Hurwitz type (24 units) and 252 of Lipschitz type (8 units). Their stabilisers in $G_2(\mathbb{Z})$ have orders 192 and 48; the pointwise stabilisers have orders 8 and 2 (the stabilisers of the zero divisors of Theorem 2.2), and in both cases the quotient acting faithfully on $\text{Im } \mathbb{H}_z$ is the octahedral group S_4 .

Proof. $1512/24 = 63$, $6048/24 = 252$; $12096/63 = 192$, $12096/252 = 48$; the pointwise stabilisers were computed directly, and $192/8 = 48/2 = 24 = |S_4|$, with S_4 the full rotation group of the cube $\{\pm i, \pm j, \pm k\}$, realised in both cases by the computation of Section 5 (orbit sizes 6, 8, 12 below). $\square$

4.2. The conic. Fix an odd prime p . The space $\text{Im } \mathbb{O} \otimes \mathbb{F}_p$ is a nondegenerate 7-dimensional quadratic space; its isotropic lines number $(p^6 - 1)/(p - 1)$ and form the maximal-parahoric level- p structures of the compact G_2 , so that the class set is $X_p = G_2(\mathbb{Z}) \backslash \{\text{isotropic lines}\}$ (Part I). For a zero divisor z , $\text{Im } \mathbb{H}_z \otimes \mathbb{F}_p$ is a nondegenerate ternary space, so its isotropic lines form a conic $C_z \cong \mathbb{P}^1(\mathbb{F}_p)$ with $p + 1$ points, on which $\text{Stab}(\mathbb{H}_z)$ acts through $S_4 \subset \text{PGL}_2(\mathbb{F}_p)$.

Theorem 4.2 (Conic law). *Let $p \geq 5$ and z a zero divisor. The $G_2(\mathbb{Z})$ -classes of the points of C_z are exactly the S_4 -orbits on $\mathbb{P}^1(\mathbb{F}_p)$, whose sizes lie in $\{6, 8, 12, 24\}$ (cube faces, vertices, edges, and generic), and the stabiliser in $G_2(\mathbb{Z})$ of a conic point in an orbit of size m has order $192/m$ (Hurwitz type) or $48/m$ (Lipschitz type). In particular each conic lies in a single class if and only if $p + 1 \in \{6, 8, 12\}$, i.e. $p \in \{5, 7, 11\}$; for $p = 3$ the conic (4 points) also lies in a single class, with the exceptional stabilisers 432 and 36.*

Lemma 4.3 (verified for $5 \leq p \leq 19$; conjectured for all $p \geq 5$). *For $p \geq 5$, an isotropic line lying on the conic of a quaternion subalgebra of given type lies on no other conic of that type.*

Proof of Theorem 4.2 from Lemma 4.3. Let $\ell \in C_z$ and $g \in G_2(\mathbb{Z})$ with $g\ell \in C_z$. Then $g\ell$ lies on the conics of $\mathbb{H}_z$ and of $g\mathbb{H}_z$, both of the type of z ; by the lemma $g\mathbb{H}_z = \mathbb{H}_z$, so $g \in \text{Stab}(\mathbb{H}_z)$. Hence the $G_2(\mathbb{Z})$ -orbit of ℓ meets C_z exactly in its $\text{Stab}(\mathbb{H}_z)$ -orbit, which is its S_4 -orbit since the pointwise stabiliser acts trivially on C_z , and the stabiliser of ℓ in $G_2(\mathbb{Z})$ equals its stabiliser in $\text{Stab}(\mathbb{H}_z)$, of order $|\text{Stab}(\mathbb{H}_z)|/m$. The orbit sizes of $S_4 \subset \text{PGL}_2(\mathbb{F}_p)$ on $\mathbb{P}^1(\mathbb{F}_p)$ are $24/|\text{point stabiliser}|$ with point stabilisers the cyclic subgroups of orders 4, 3, 2, 1, giving 6, 8, 12, 24; S_4 is transitive on $\mathbb{P}^1(\mathbb{F}_p)$ iff $p + 1 \in \{6, 8, 12\}$ (the case 4 being $p = 3$, where the lemma fails and the stabilisers are larger). $\square$

Computation 4.4. Exact orbit computations for $p = 3, 5, 7, 11, 13, 17, 19$ (four to forty zero divisors of each type per prime) give the class structure of the conic points shown in Table 1; the stabiliser identity $|\text{Stab}(\ell)| = |\text{Stab}(\mathbb{H}_z)|/m$ holds in every case with $p \geq 5$, which is equivalent to Lemma 4.3 for those primes. The classes touched were also matched against the full class sets X_5 (3 classes, stabilisers 32, 8, 6: the 6-class is touched by no zero divisor) and X_7 (8 classes).

Remark 4.5. Lemma 4.3 is a statement about the Coxeter order: two distinct quaternion suborders of the same type cannot share an isotropic line modulo $p \geq 5$. The incidence counts make it plausible in general ($63(p + 1)$ and $252(p + 1)$ incidences onto classes of the predicted sizes), and we expect a proof from the fact that the relevant indices and discriminants involve only the primes 2 and 3; we have not written it.

5. ZERO-DIVISOR PERIODS

5.1. Definition. Let f be an algebraic modular form of weight π for the compact G_2 at a parahoric level at $p = 3$: a function on the level-3 structures x (isotropic lines, planes or flags

p	$p + 1$	Lipschitz: (points, stabiliser)	Hurwitz: (points, stabiliser)
3	4	(4, 36)	(4, 432)
5	6	(6, 8)	(6, 32)
7	8	(8, 6)	(8, 24)
11	12	(12, 4)	(12, 16)
13	14	(8, 6) + (6, 8)	(8, 24) + (6, 32)
17	18	(12, 4) + (6, 8)	(12, 16) + (6, 32)
19	20	(12, 4) + (8, 6)	(12, 16) + (8, 24)

TABLE 1. Class structure of the conic points of a zero divisor; stabilisers are $48/m$ and $192/m$ for $p \geq 5$.

mod 3) with $f(x\gamma) = \pi(\gamma)^{-1}f(x)$, in the conventions of [2]. For a zero divisor z with continuous stabiliser K_z , and $u \in \pi^{K_z}$, put

$$P_{z,u}(f) = \sum_{x \text{ compatible with } z} \langle f(x), u \rangle, \quad x \text{ compatible} \iff \text{the line of } x \text{ lies in } \text{Im } \mathbb{H}_z \text{ mod } 3,$$

and more generally, for any orbit O of the finite group $\text{Stab}(\mathbb{H}_z)$ on the structures, $P_{z,O}(f) = \text{proj}_{\pi^{K_z}} \sum_{x \in O} f(x)$. Every zero divisor has exactly 4 compatible lines (the conic at $p = 3$).

5.2. Trivial weight. At the maximal parahoric level X_3 has two classes (stabilisers 432, 36) and T_2 -eigenvalues 126 (Eisenstein) and 9 (the lift of 3.6.a.a). By Theorem 4.2 the four conic points of a Hurwitz zero divisor lie in the 432-class and those of a Lipschitz one in the 36-class, so $P_z(f) = 4f(\text{class of } z)$: for the cusp form, normalised $f = (-1, 1/12)$, the periods are -4 and $1/3$.

5.3. Weights 14 and 27. Weight 14 (adjoint): one form at level 3 ($T_2 = -18$, at the plane and flag levels), whose conic periods vanish for both types. Weight 27 ($2\omega_1$): at the Iwahori level a 7-dimensional space with $T_2 \in \{40.5 (\times 2), 0 (\times 3), -10.768, 6.268\}$, the irrational pair being the roots of $x^2 + 4.5x - 67.5$ and new at the Iwahori level; the others descend to the maximal parahoric levels, where $(T_2, T_5) = (81/2, 2034/5)$ and $(0, 11466/25)$ (twice), computed with the calibration (9, 810) and (126, 19530) in trivial weight.

Computation 5.1. (i) The conic period $P_{z,u}$ vanishes identically on the Iwahori-new forms and is nonzero on all old forms, for both types. (ii) Over all $\text{Stab}(\mathbb{H}_z)$ -orbits (51 orbits for Lipschitz, 21 for Hurwitz type), the Iwahori-new forms have nonzero $P_{z,O}$ on 41/51 resp. 16/21 orbits, vanishing exactly on the orbits containing the conic flags (and a few others); the old forms are nonzero on every orbit.

5.4. Interpretation. In [5] the pair $D^1 \subset \text{Aut}(\mathbb{O})$ sits in a see-saw inside the adjoint E_7 of the Albert algebra, with $\text{Aut}(\mathbb{O}) \times \text{PGSp}_6$ and $D^1 \times G_D$ (G_D of type D_6) as dual pairs; the theta lift of the trivial representation of D^1 is a residue of a Siegel Eisenstein series on G_D , and by Pollack's integral and Chenevier's identity $L(s, \pi, \text{Spin}) = \zeta(s)L(s, \pi', \text{Std})$ for G_2 -parameters, the D^1 -period of a form on the compact G_2 is governed by $L(1, \pi', \text{Std}) \neq 0$. Our periods are D^1 -periods with specific local data at 3. Computation 5.1(ii) is in accordance: the genuine (Iwahori-new) forms have nonzero D^1 -periods, and what vanishes is the period *along the conic*, a local statement about Iwahori-fixed vectors at $p = 3$. The non-integral eigenvalues of the old forms, with denominators 2, 5, 25, and their multiplicities suggest endoscopic parameters through $\text{SO}(4) \subset G_2$, the stabiliser of a quaternion subalgebra; the vector-valued Satake dictionary

needed to decide this is the next task. No level-one forms exist in weights 14, 27, 77' ($G_2(\mathbb{Z})$ -invariants vanish), so the old forms are genuinely level 3.

6. NEGATIVES AND BOUNDARIES

6.1. Sedenion spectral geometries. Farnsworth and Boyle [7, 8] extend Connes' spectral triples to non-associative algebras: the Hilbert space must be a bimodule in the algebra's own class (every identity surviving the replacement of one algebra element by a vector), the order conditions become vanishing associators in $B = \Omega A \oplus H$, and the gauge group is $\text{Aut}(A)$. For the sedenions acting on themselves: the flexible order-zero law holds and the alternative ones fail; the derivation algebra is 14-dimensional and inner (inside $\mathfrak{so}(16) \oplus \mathbb{R}$); the fermions decompose as $2(1+7)$ under G_2 with the S_3 of $\text{Aut}(\mathbb{S}) = G_2 \times S_3$ as a family symmetry; the commutant of the left action is trivial, forcing $\gamma = \pm 1$ and $D_F = 0$. On two copies $\mathbb{S} \oplus \mathbb{S}$ with $\gamma = \text{diag}(1, -1)$, the flexible order-one condition forces D_F to be a multiple of the identity for either real structure; with the opposite bimodule on the second copy, no D_F survives; on the complex sedenions $\mathbb{C} \otimes \mathbb{S}$ with real-linear D_F , only complex scalars survive. The sedenion analogue $H_3(\mathbb{S})$ of the Albert algebra violates the Jordan identity (relative size 0.96; $H_3(\mathbb{O})$ satisfies it to 10^{-15}), so the Jordan-geometry route [8] excludes the sedenions and admits the Albert algebra. In every construction the zero divisors do not enter the action.

6.2. Euler products along the ladder. The norm-counting Dirichlet series $\sum r(n)n^{-s}$ has an Euler product iff r is multiplicative. It does for $\mathbb{Z}$, $\mathbb{Z}[i]$, the Hurwitz quaternions (D_4), the Coxeter octonions (E_8 , $r = 240\sigma_3$) and the integral sedenions ($E_8 \oplus E_8$, $r = 480\sigma_7$) — the last only because $M_8(\text{SL}_2(\mathbb{Z}))$ has no cusp forms — and fails for the integral 32-ions (E_8^4), since $E_4^4 = E_{16} + \frac{3456000}{3617} E_4 \Delta$. The counting zeta function of the 32-ions, $\frac{16320}{3617} \zeta(s) \zeta(s-15) + c L(E_4 \Delta, s)$, has two zeros on its critical line $\Re s = 8$ below $t \approx 15$ and none up to $t = 100$, where the Eisenstein part dominates by 10^5 – 10^6 ; the Leech lattice's $\zeta(s) \zeta(s-11) - L(\Delta, s)$ has no zeros at all on $\Re s = 6$ to $t = 120$, its 57 zeros to height 198 lying near $\Re s \in [11.1, 12.5]$ with spacings more rigid than GUE. The product over primes is lost exactly where Δ enters.

7. OPEN PROBLEMS

- (1) Prove Lemma 4.3 (two quaternion suborders of the same type share no isotropic line mod $p \geq 5$).
- (2) The vector-valued Satake dictionary for weight $2\omega_1$ and the identification of the old forms $(81/2, 2034/5)$, $(0, 11466/25)$; their exceptional theta lifts to genus-3 Siegel forms of level 3.
- (3) The local analysis of the conic period: is it a $D^1(\mathbb{Q}_3)$ -distinction statement for Iwahori-new local representations of $G_2(\mathbb{Q}_3)$?
- (4) Index the seven free $G_2(\mathbb{Z})$ -orbits on zero-divisor pairs; the 32-ions; the Okubo integral orders of [11] and their idempotent geometry against the conic law.
- (5) Part III: the J_E half of the F_4 class set at level 5 (88% of the mass found; exact deficits 720799/96768 and 3).

APPENDIX A. CODE AND DATA

Repository `chrisyounge-create/Cayley-Agent-Repository`: `sedenion.py` (doubling, laws), `sedenion_geometry.py`, `sedenion_step2.py`, `sedenion_step3.py`, `sedenion_step3_complex.py` (spectral geometries), `zd_hecke.py` (stabiliser, annihilator, multiplicities), `g2table/zd_periods.py`, `g2table/zd_periods_vec.py` (periods), `beurling.py` (prime-correlation model). Dropbox notes `RH/SEDENIONS.md` through `RH/NEXT_RUN_ITEMS.1_TO.4.md` contain the session records.

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